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Java SE 27 & JDK 27

Class OptionalDouble

java.lang.Object
    java.util.OptionalDouble

```
public final class OptionalDouble
extends Object
```

A container object which may or may not contain a `double` value. If a value is present, `isPresent()` returns `true`. If no value is present, the object is considered *empty* and `isPresent()` returns `false`.

Additional methods that depend on the presence or absence of a contained value are provided, such as `orElse()` (returns a default value if no value is present) and `ifPresent()` (performs an action if a value is present).

This is a value-based class; programmers should treat instances that are equal as interchangeable and should not use instances for synchronization, or unpredictable behavior may occur. For example, in a future release, synchronization may fail.

API Note:

`OptionalDouble` is primarily intended for use as a method return type where there is a clear need to represent "no result." A variable whose type is `OptionalDouble` should never itself be `null`; it should always point to an `OptionalDouble` instance.

Since:

1.8

Method Summary

All Methods	Static Methods	

Modifier and Type	Method	Description
static OptionalDouble	**empty**()	Returns an empty `OptionalDouble` instance.
boolean	**equals**(Object obj)	Indicates whether some other object is "equal to" this `OptionalDouble`.
double	**getAsDouble**()	If a value is present, returns the value, otherwise throws `NoSuchElementException`.

int	**hashCode**()	Returns the hash code of the value, if present, otherwise `0` (zero) if no value is present.
void	**ifPresent**(DoubleConsumer action)	If a value is present, performs the given action with the value, otherwise does nothing.
void	**ifPresentOrElse**(DoubleConsumer action, Runnable emptyAction)	If a value is present, performs the given action with the value, otherwise performs the given empty-based action.
boolean	**isEmpty**()	If a value is not present, returns `true`, otherwise `false`.
boolean	**isPresent**()	If a value is present, returns `true`, otherwise `false`.
static OptionalDouble	**of**(double value)	Returns an `OptionalDouble` describing the given value.
double	**orElse**(double other)	If a value is present, returns the value, otherwise returns `other`.
double	**orElseGet**(DoubleSupplier supplier	If a value is present, returns the value, otherwise returns the result produced by the supplying function.
double	**orElseThrow**()	If a value is present, returns the value, otherwise throws `NoSuchElementException`.

<X extends Throwable> double	**orElseThrow**(Supplier<? extends X> exceptionSupplier)	If a value is present, returns the value, otherwise throws an exception produced by the exception supplying function.
DoubleStream	**stream**()	If a value is present, returns a sequential DoubleStream containing only that value, otherwise returns an empty DoubleStream.
String	**toString**()	Returns a non-empty string representation of this OptionalDouble suitable for debugging.

Methods declared in class Object

clone, finalize, getClass, notify, notifyAll, wait, wait, wait

Method Details

empty

```
public static OptionalDouble empty()
```

Returns an empty OptionalDouble instance. No value is present for this OptionalDouble.

API Note:

Though it may be tempting to do so, avoid testing if an object is empty by comparing with == or != against instances returned by OptionalDouble.empty(). There is no guarantee that it is a singleton. Instead, use isEmpty() or isPresent().

Returns:

an empty OptionalDouble.

of

```
public static OptionalDouble of(double value)
```

Returns an OptionalDouble describing the given value.

Parameters:

`value` - the value to describe

Returns:

an `OptionalDouble` with the value present

getAsDouble

```
public double getAsDouble()
```

If a value is present, returns the value, otherwise throws `NoSuchElementException`.

API Note:

The preferred alternative to this method is `orElseThrow()`.

Returns:

the value described by this `OptionalDouble`

Throws:

`NoSuchElementException` - if no value is present

isPresent

```
public boolean isPresent()
```

If a value is present, returns `true`, otherwise `false`.

Returns:

`true` if a value is present, otherwise `false`

isEmpty

```
public boolean isEmpty()
```

If a value is not present, returns `true`, otherwise `false`.

Returns:

`true` if a value is not present, otherwise `false`

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ifPresent

```
public void ifPresent(DoubleConsumer action)
```

If a value is present, performs the given action with the value, otherwise does nothing.

Parameters:

`action` - the action to be performed, if a value is present

Throws:

`NullPointerException` - if value is present and the given action is `null`

ifPresentOrElse

```
public void ifPresentOrElse(DoubleConsumer action,
                            Runnable emptyAction)
```

If a value is present, performs the given action with the value, otherwise performs the given empty-based action.

Parameters:

`action` - the action to be performed, if a value is present

`emptyAction` - the empty-based action to be performed, if no value is present

Throws:

`NullPointerException` - if a value is present and the given action is `null`, or no value is present and the given empty-based action is `null`.

Since:

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stream

```
public DoubleStream stream()
```

If a value is present, returns a sequential `DoubleStream` containing only that value, otherwise returns an empty `DoubleStream`.

API Note:

This method can be used to transform a `Stream` of optional doubles to a `DoubleStream` of present doubles:

```
Stream<OptionalDouble> os = ..
DoubleStream s = os.flatMapToDouble(OptionalDouble::stream)
```

Returns:

the optional value as a `DoubleStream`

Since:

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orElse

```
public double orElse(double other)
```

If a value is present, returns the value, otherwise returns `other`.

Parameters:

`other` - the value to be returned, if no value is present

Returns:

the value, if present, otherwise `other`

orElseGet

```
public double orElseGet(DoubleSupplier supplier)
```

If a value is present, returns the value, otherwise returns the result produced by the supplying function.

Parameters:

`supplier` - the supplying function that produces a value to be returned

Returns:

the value, if present, otherwise the result produced by the supplying function

Throws:

`NullPointerException` - if no value is present and the supplying function is `null`

orElseThrow

```
public double orElseThrow()
```

If a value is present, returns the value, otherwise throws `NoSuchElementException`.

Returns:

the value described by this `OptionalDouble`

Throws:

`NoSuchElementException` - if no value is present

Since:

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orElseThrow

```
public <X extends Throwable> double orElseThrow
(Supplier<? extends X> exceptionSupplier)
                                          throws X
```

If a value is present, returns the value, otherwise throws an exception produced by the exception supplying function.

API Note:

A method reference to the exception constructor with an empty argument list can be used as the supplier. For example, `IllegalStateException::new`

Type Parameters:

`X` - Type of the exception to be thrown

Parameters:

`exceptionSupplier` - the supplying function that produces an exception to be thrown

Returns:

the value, if present

Throws:

`X` - if no value is present

`NullPointerException` - if no value is present and the exception supplying function is `null`

equals

```
public boolean equals(Object obj)
```

Indicates whether some other object is "equal to" this `OptionalDouble`. The other object is considered equal if:

- it is also an `OptionalDouble` and;
- both instances have no value present or;
- the present values are "equal to" each other via `Double.compare() == 0`.

Overrides:

`equals` in class `Object`

Parameters:

`obj` - an object to be tested for equality

Returns:

`true` if the other object is "equal to" this object otherwise `false`

See Also:

`Object.hashCode()`, `HashMap`

hashCode

```
public int hashCode()
```

Returns the hash code of the value, if present, otherwise `0` (zero) if no value is present.

Overrides:

hashCode in class Object

Returns:

hash code value of the present value or `0` if no value is present

See Also:

Object.equals(java.lang.Object),
System.identityHashCode(Object)

toString

```
public String toString()
```

Returns a non-empty string representation of this `OptionalDouble` suitable for debugging. The exact presentation format is unspecified and may vary between implementations and versions.

Overrides:

toString in class Object

Implementation Requirements:

If a value is present the result must include its string representation in the result. Empty and present `OptionalDouble`s must be unambiguously differentiable.

Returns:

the string representation of this instance

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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